



Facultad de
Ciencias
UNAM

AVANCES DE TESIS DE LICENCIATURA

Cálculo numérico de propiedades ópticas de eritrocitos sanos y enfermos

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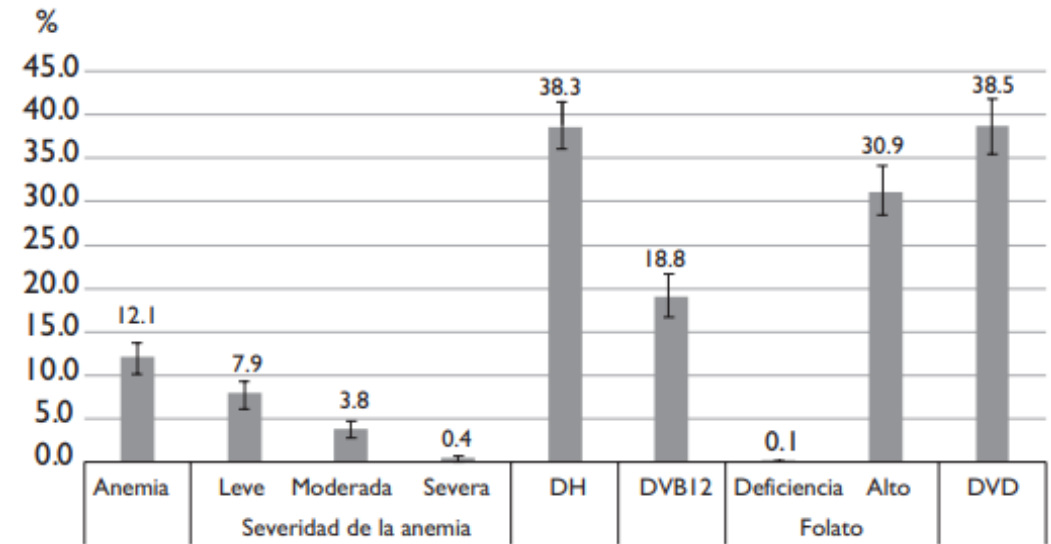
M. en C. Jonathan Alexis Urrutia Anguiano
Dr. Alejandro Reyes Coronado



Motivación

Anemia en México

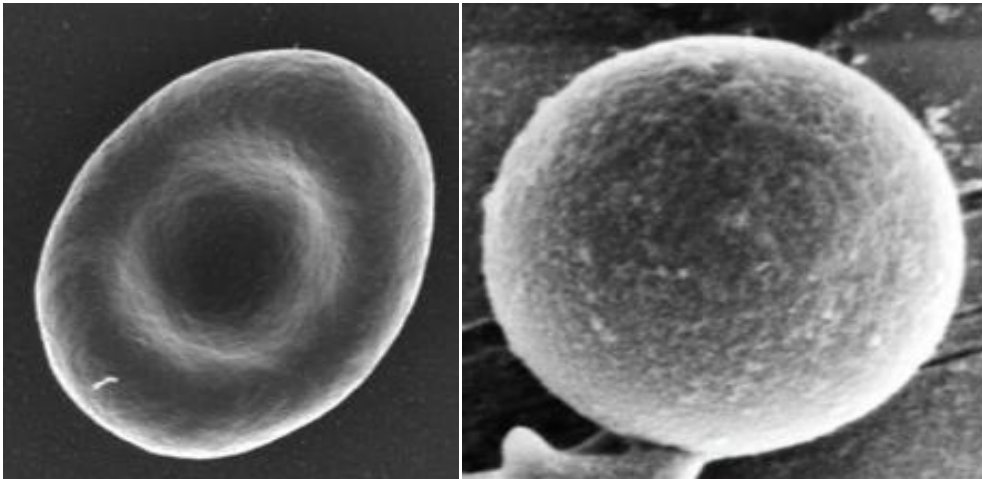
- En México, la prevalencia de anemia en los últimos 4 años alcanza el 12.1%
- La **anemia** es una disminución de la capacidad de transporte de oxígeno de la sangre. Se clasifica según:
 - Morfología
 - Composición



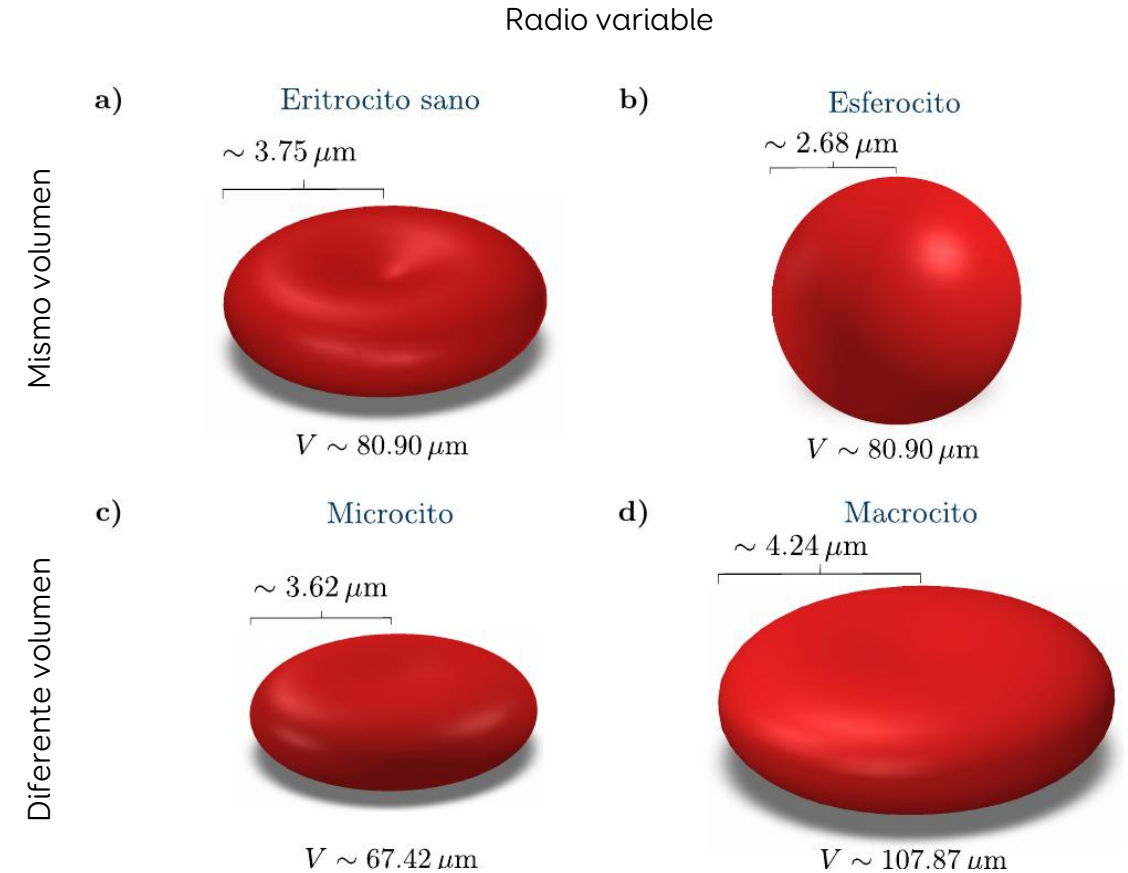
Shamah-Levy, T., et al. *Salud Pública de México*, 67(6), 2025

Motivación

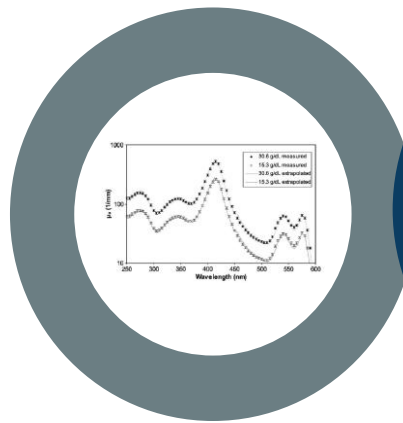
Morfología de eritrocitos sanos y enfermos



Alummoottil, M., et al. *Hemato*, **15**(4), 2010
Khanna, R., et al. *Proc. Natl. Acad. Sci.*, **99**(10), 2002



Plan de trabajo



Construcción de funciones dieléctricas de eritrocitos



Comparación de CST vs Mie



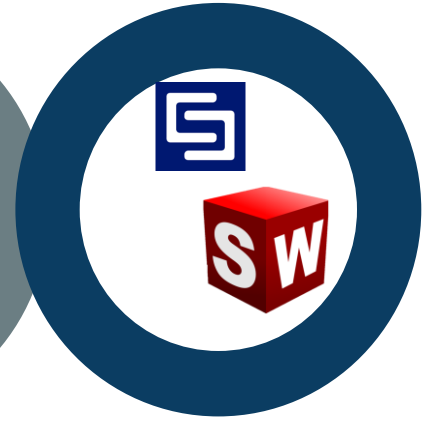
Análisis de convergencia numérica en CST



Evaluación del efecto del mallado



Construcción de geometrías de eritrocitos



Cálculo de propiedades ópticas de eritrocitos

0%



Preámbulo

90%



Teoría

0%

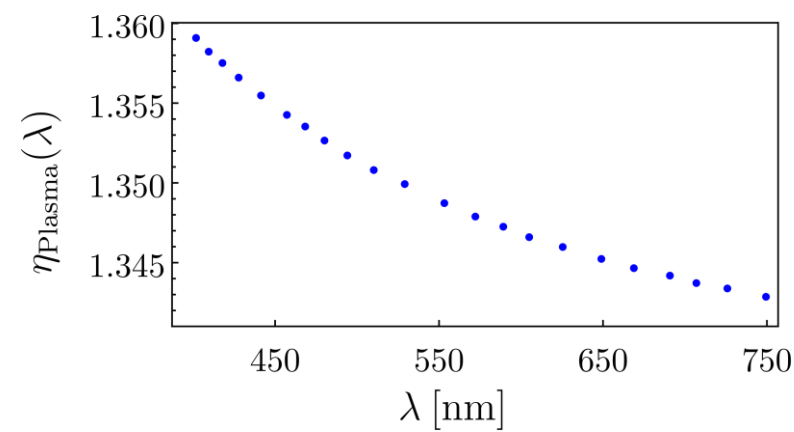


Resultados y conclusiones

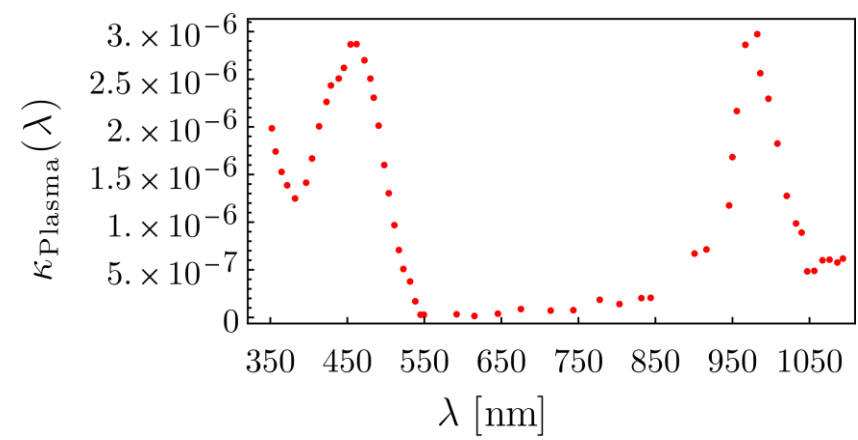
Datos experimentales

$$n(\lambda) = \eta(\lambda) + i\kappa(\lambda)$$

Plasma



Liu, S., et al. *J. Biomed. Opt.*, **24**(3), 2019



Meinke, M., et al. *J. Biomed. Opt.*, **12**(1), 2007

Eritrocitos

Parte real:

Rango espectral
250 - 1100 nm

$$\eta_{\text{Hb}}(\lambda, c_{\text{Hb}}) = \eta_{\text{H}_2\text{O}}(\lambda)[\beta(\lambda)c_{\text{Hb}} + 1]$$

Friebel, M. y Meinke, M. *Appl. Opt.*, **45**(12), 2006

Parte imaginaria:

Concentración	$\kappa_{\text{Hb}}(\lambda)$	Rango espectral
• 28.7 g/dL	Microcitosis	250 - 600 nm
• 30.6 g/dL	Sano	250 - 1050 nm
• 34 g/dL	Macrocitosis/ esferocitosis	250 - 600 nm

Friebel, M. y Meinke, M. *Appl. Opt.*, **10**(6), 2005

Relaciones

1

Obtención de la parte real e imaginaria de la función dieléctrica mediante las relaciones

$$\begin{aligned}\operatorname{Re}\{\varepsilon\} &= \eta^2 - \kappa^2 \\ \operatorname{Im}\{\varepsilon\} &= 2\eta\kappa\end{aligned}$$

2

Ajuste de lorentzianas a la parte imaginaria de la función dieléctrica

$$\operatorname{Im}\{\varepsilon\} = \sum_{i=1}^N \frac{A_i \gamma_i \omega}{(\omega_{0_i}^2 - \omega^2)^2 + \gamma_i^2 \omega^2}$$

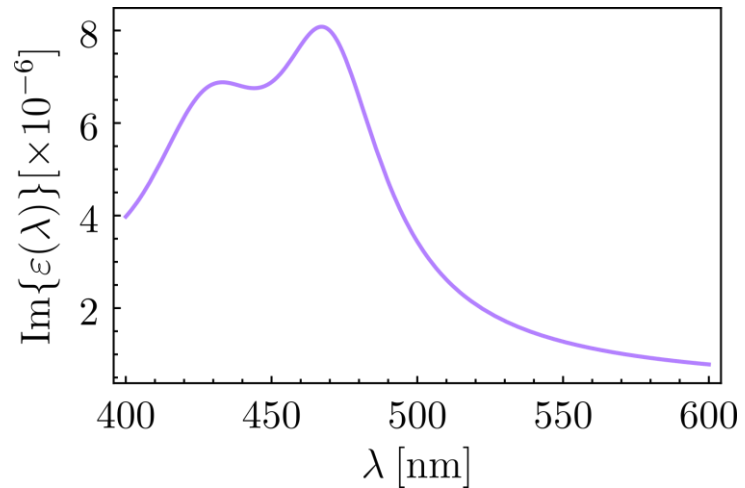
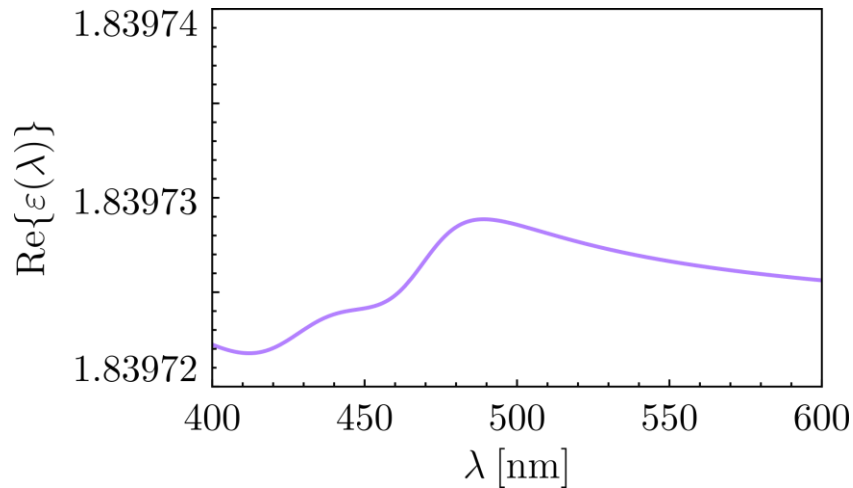
3

Aplicación de las relaciones sustractivas de Kramers-Kronig (SSKK)

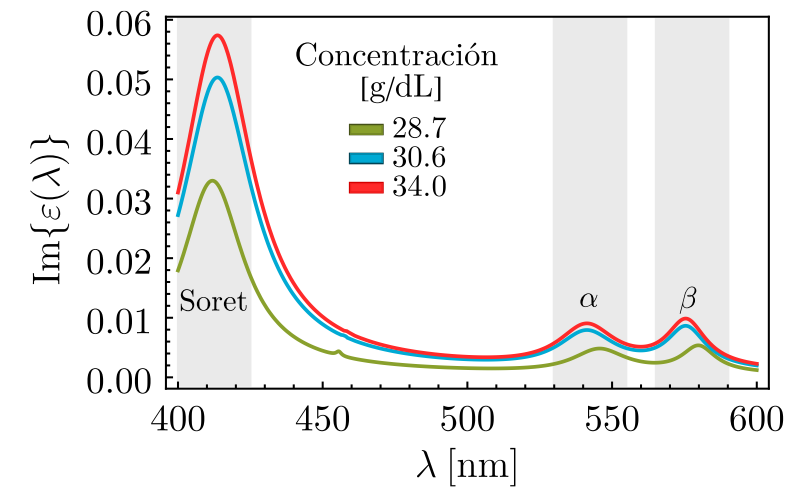
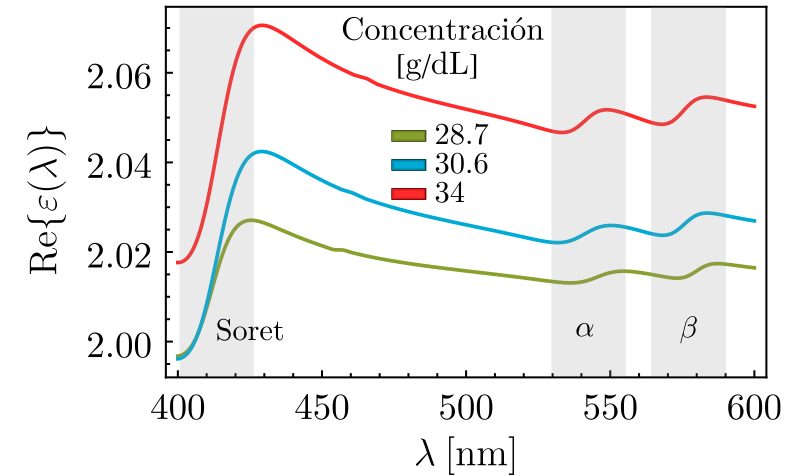
$$\begin{aligned}\operatorname{Re}\{\varepsilon\}(\omega) &= \operatorname{Re}\{\varepsilon\}(\omega_1) + \frac{2(\omega^2 - \omega_1^2)}{\pi} \operatorname{PV} \int_0^\infty \frac{\omega' \operatorname{Im}\{\varepsilon\}(\omega')}{(\omega'^2 - \omega^2)(\omega'^2 - \omega_1^2)} d\omega' \\ \operatorname{Im}\{\varepsilon\}(\omega) &= \operatorname{Im}\{\varepsilon\}(\omega_1) - \frac{2}{\pi} (\omega - \omega_1) \operatorname{PV} \int_0^\infty \frac{(\operatorname{Re}\{\varepsilon\}(\omega') - 1)(\omega'^2 + \omega\omega_1)}{(\omega'^2 - \omega^2)(\omega'^2 - \omega_1^2)} d\omega'\end{aligned}$$

Eritrocitos y plasma

Plasma

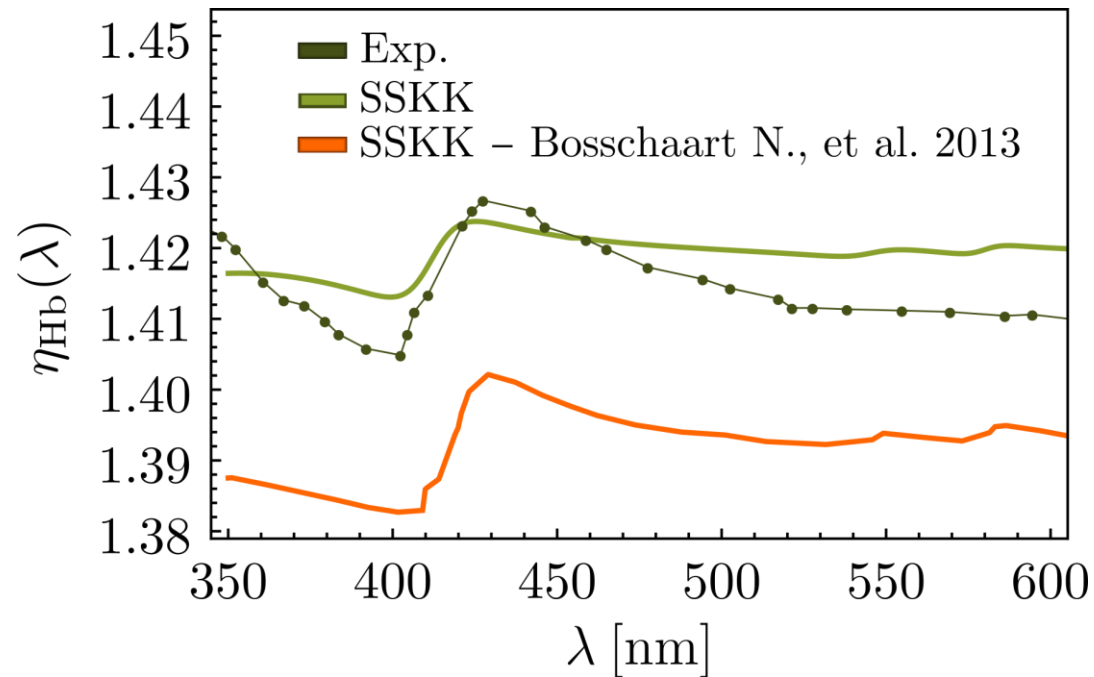


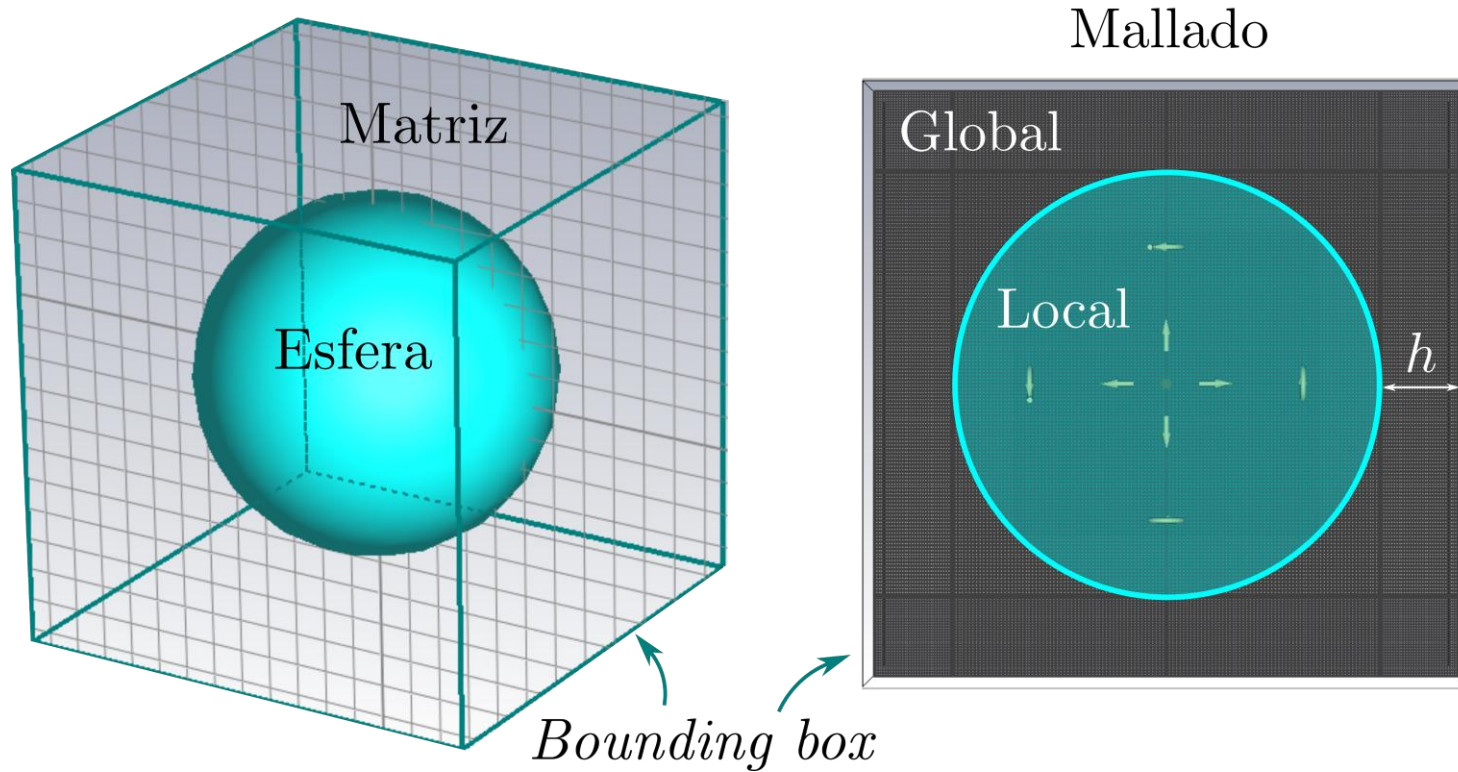
Eritrocitos



Eritrocitos y plasma

Comparación de los resultados obtenidos con los reportados por Bosschaart N, et al. 2013



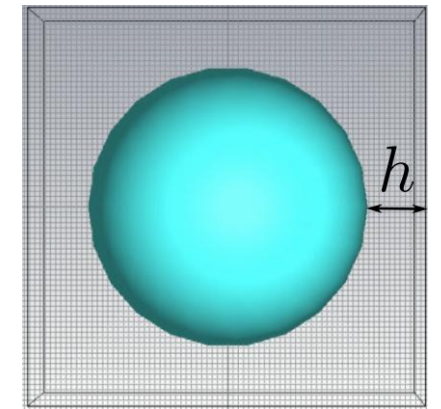
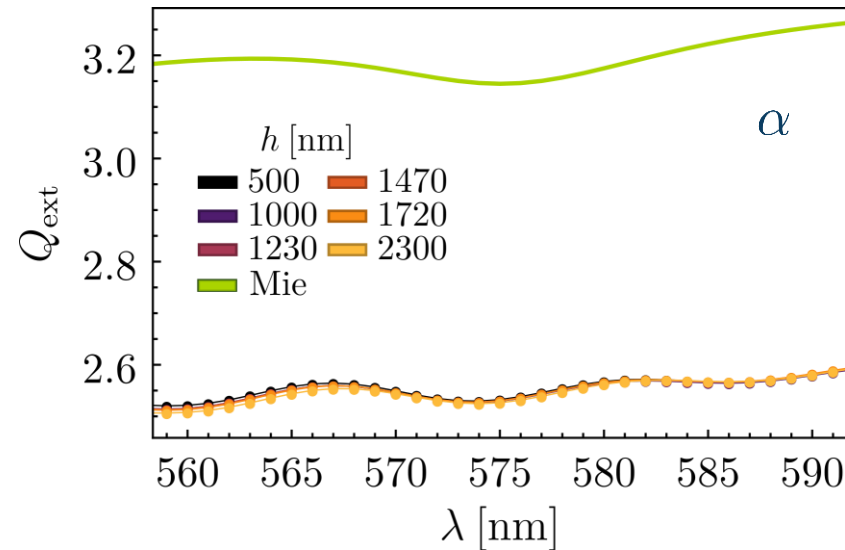
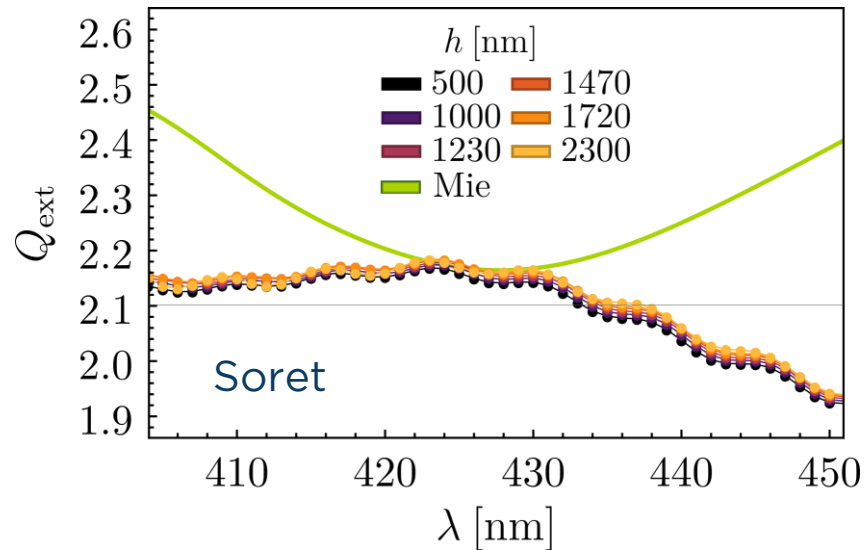
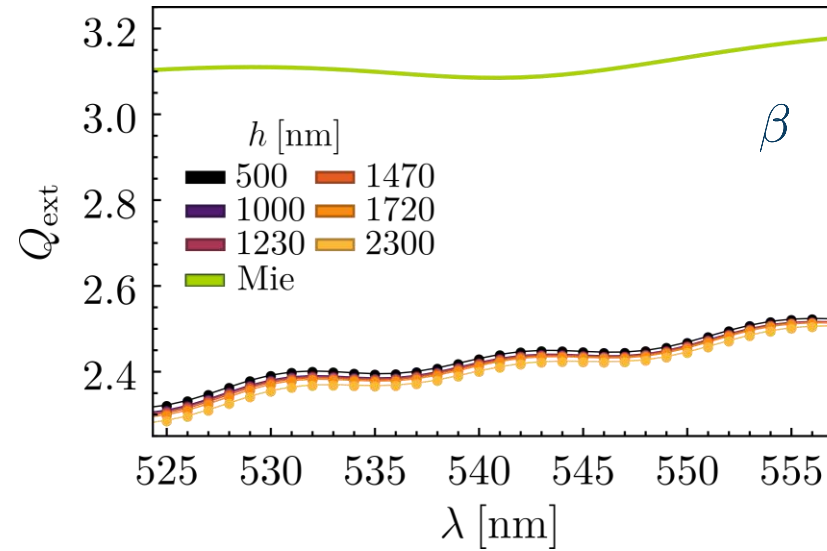
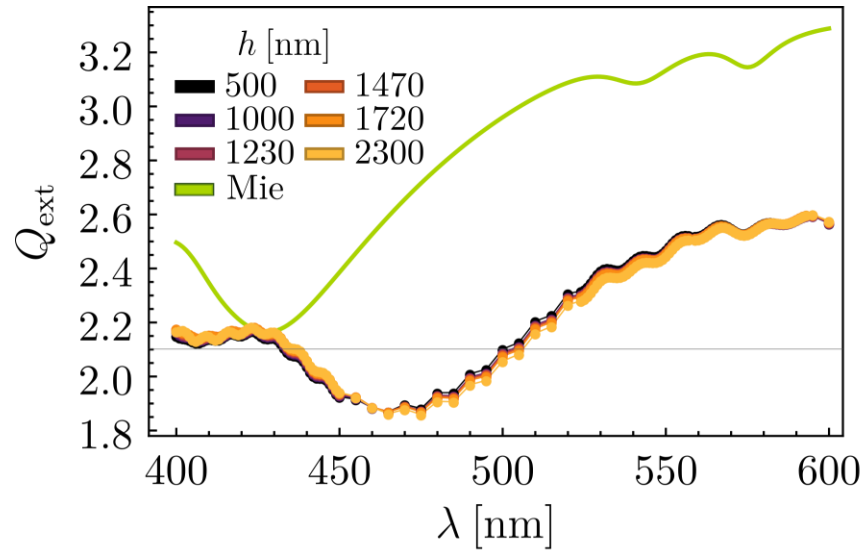


Parámetros

- Distancia entre *bounding box* a la superficie de la esfera (h)
- Mallado global
- Mallado local
 - Superficie
 - Volumen
- Estimated reflection level (Perfect Matching Layer - PML)

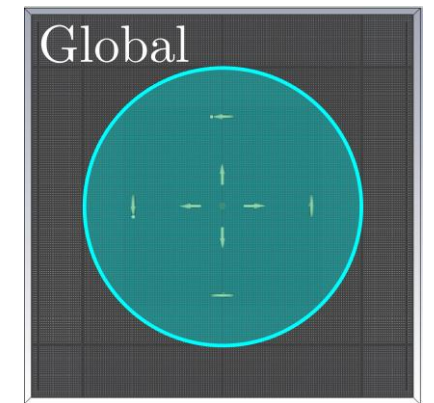
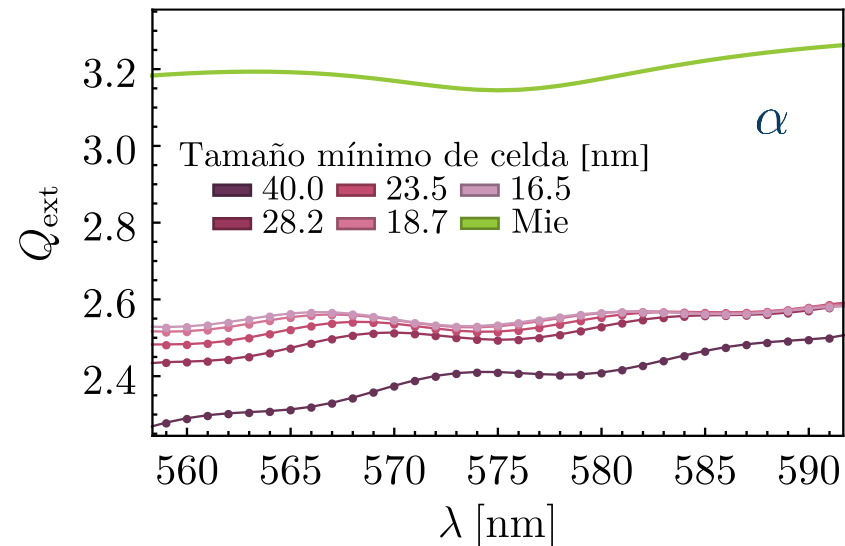
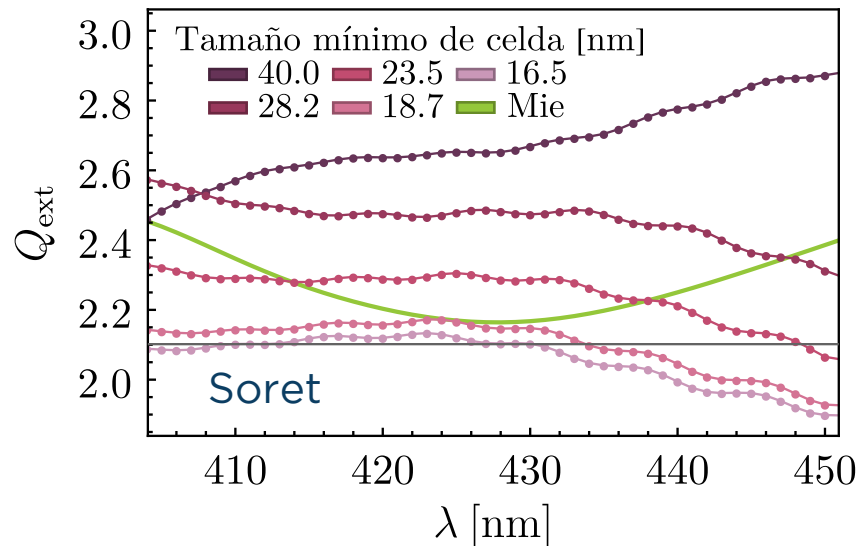
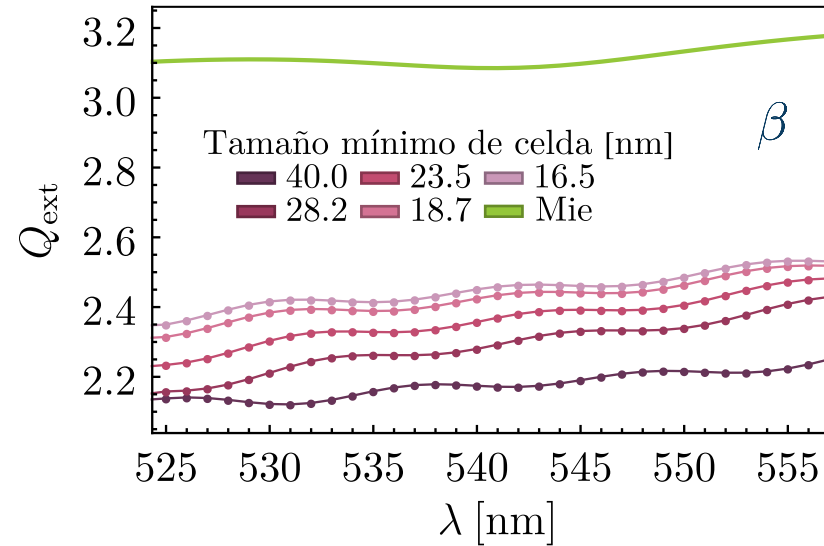
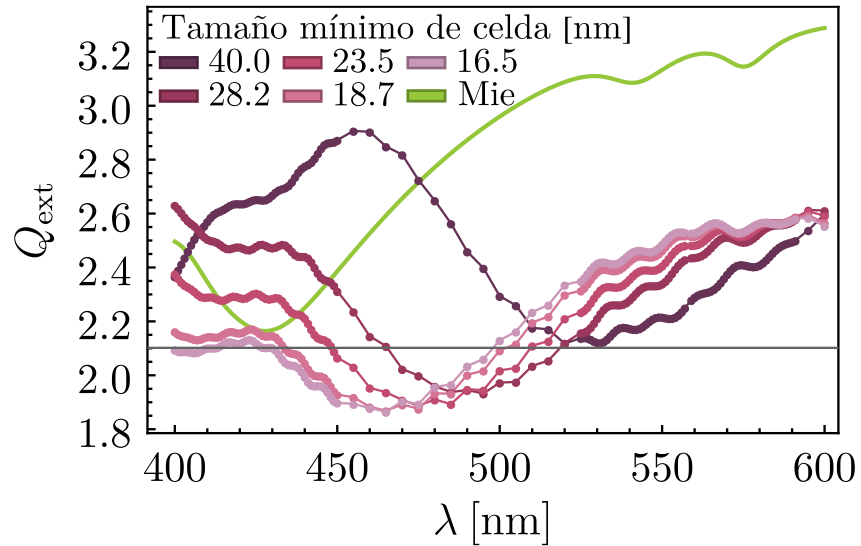
CST Studio vs Mie

Distancia h



CST Studio vs Mie

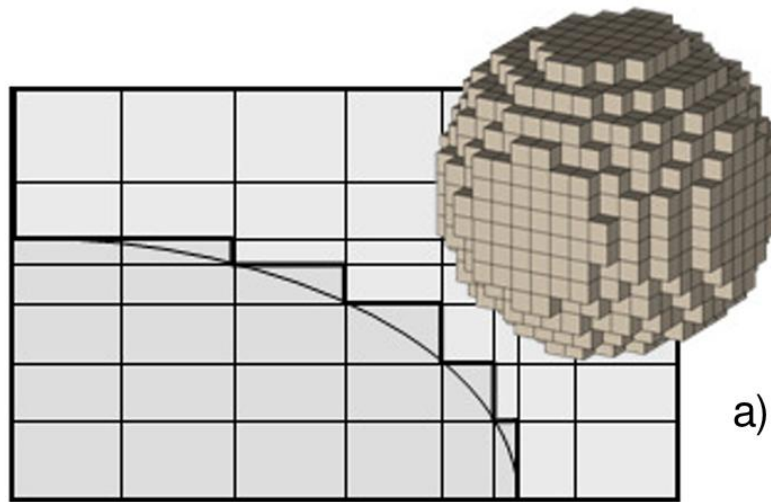
Mallado global



- No se encontraron diferencias al refinar el mallado en la superficie de la esfera

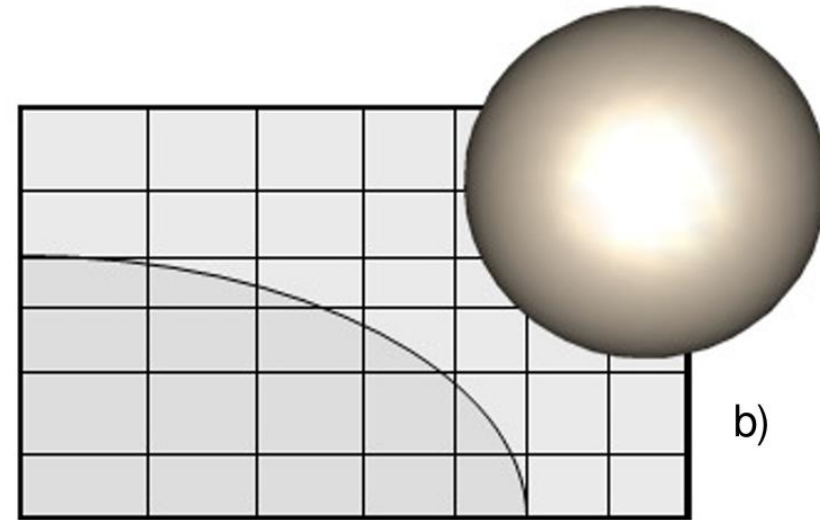


*Perfect Boundary Approximation
(PBA)*



a)

Staircase Approximation

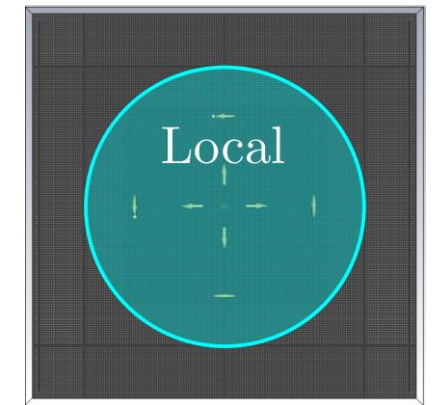
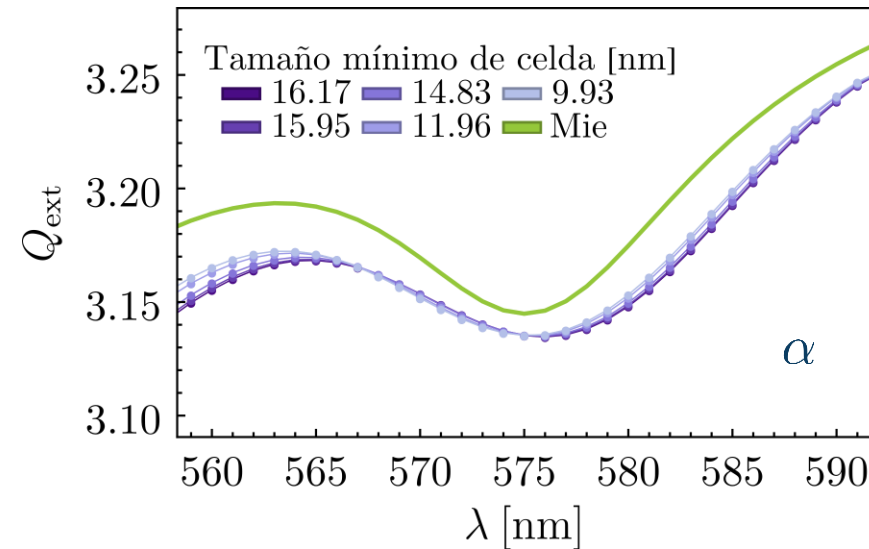
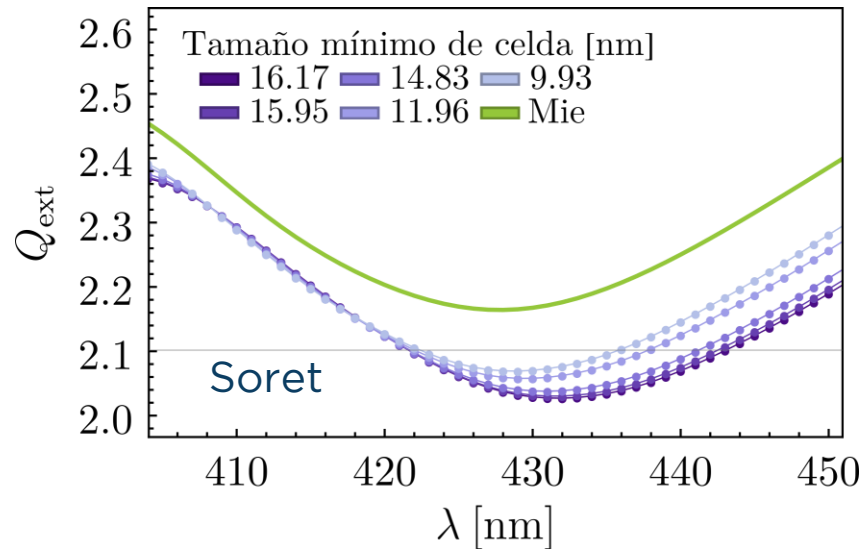
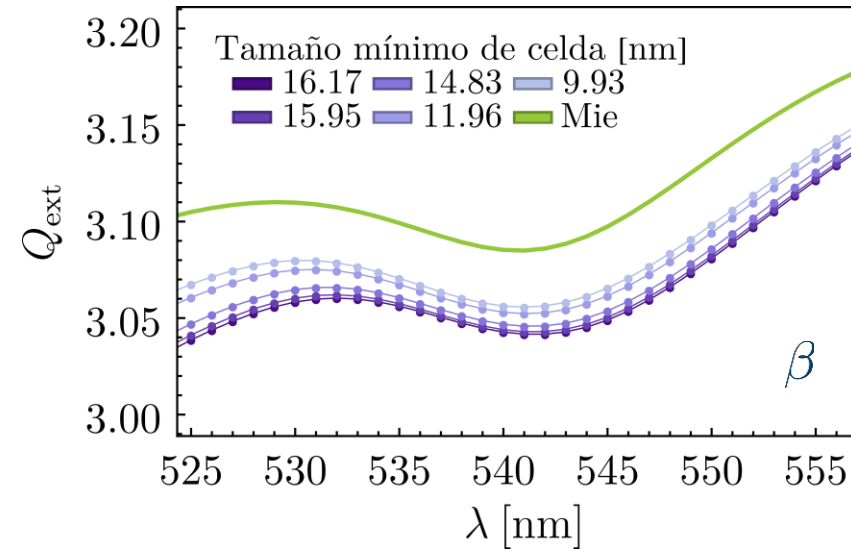
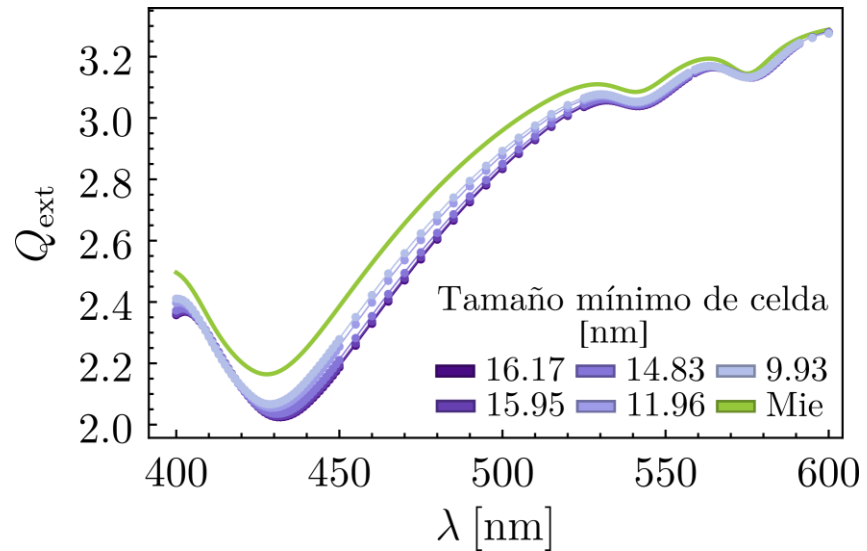


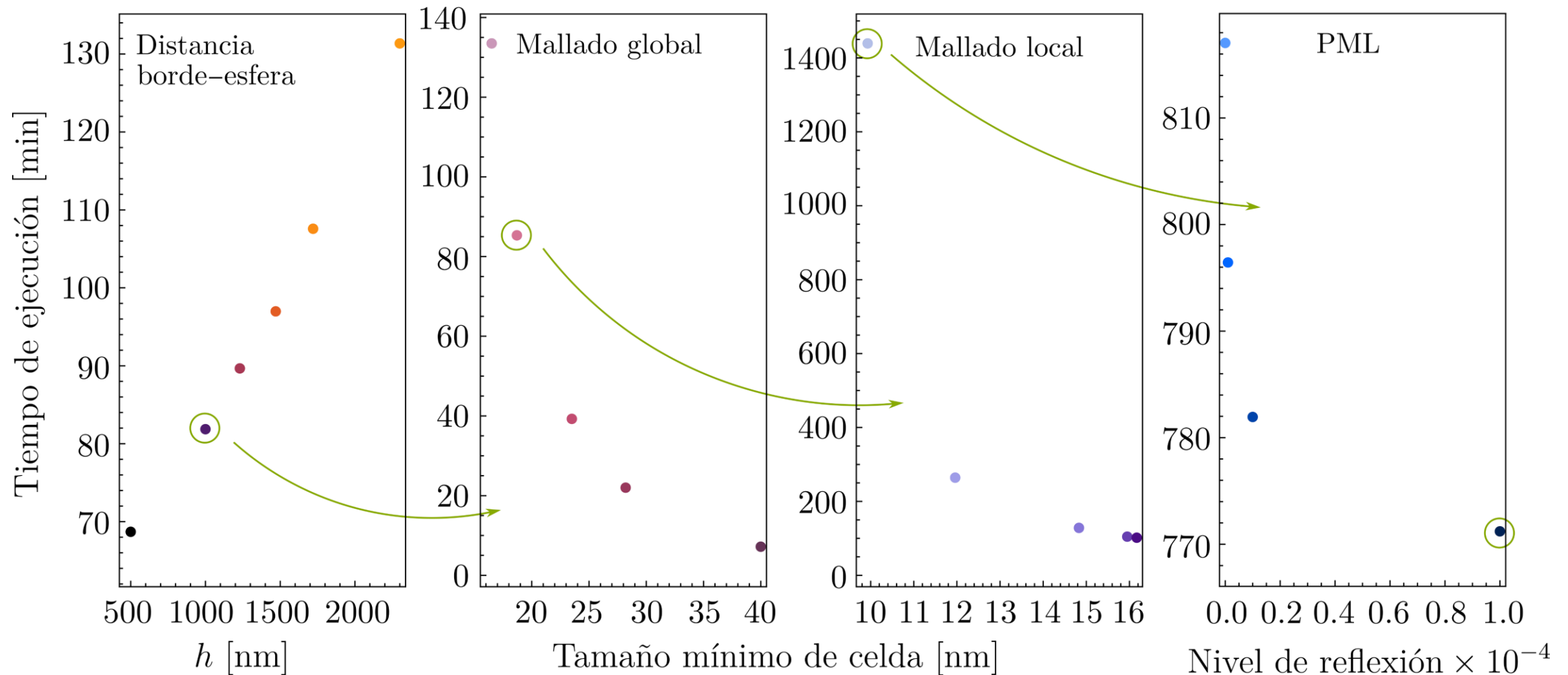
b)

PBA

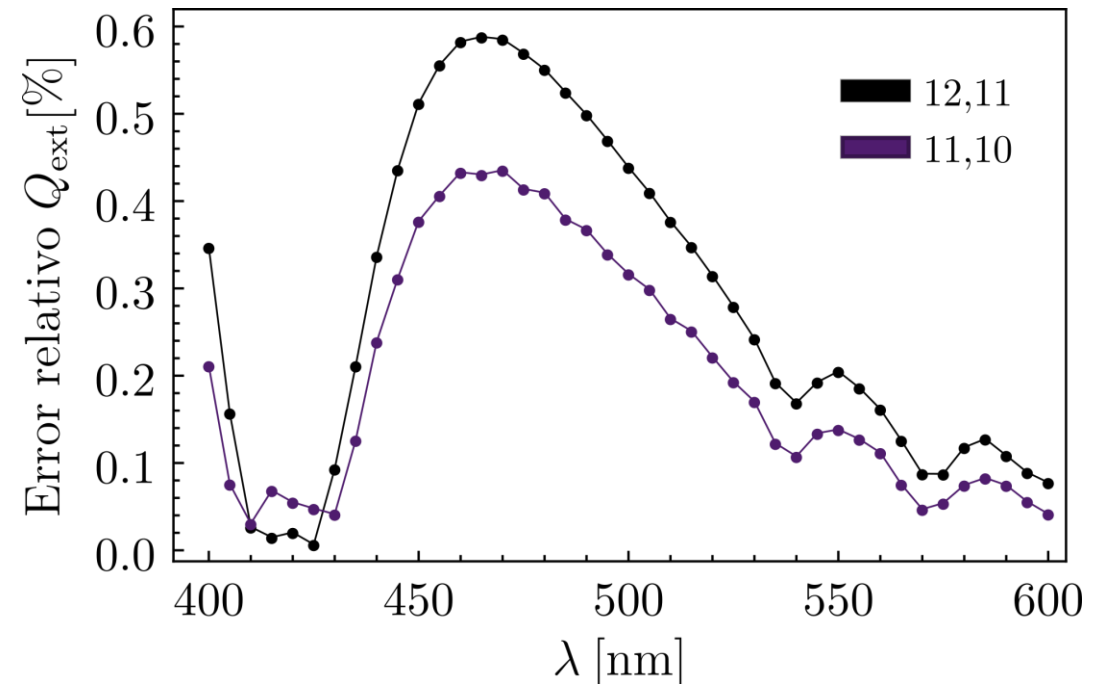
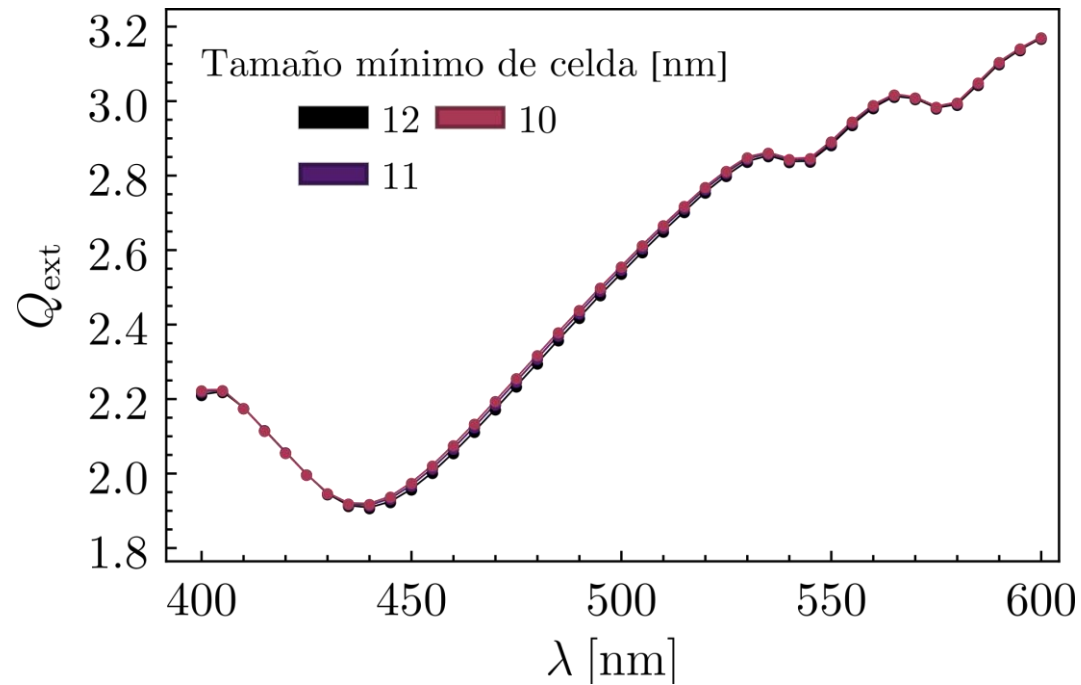
CST Studio vs Mie

Mallado local: volumen



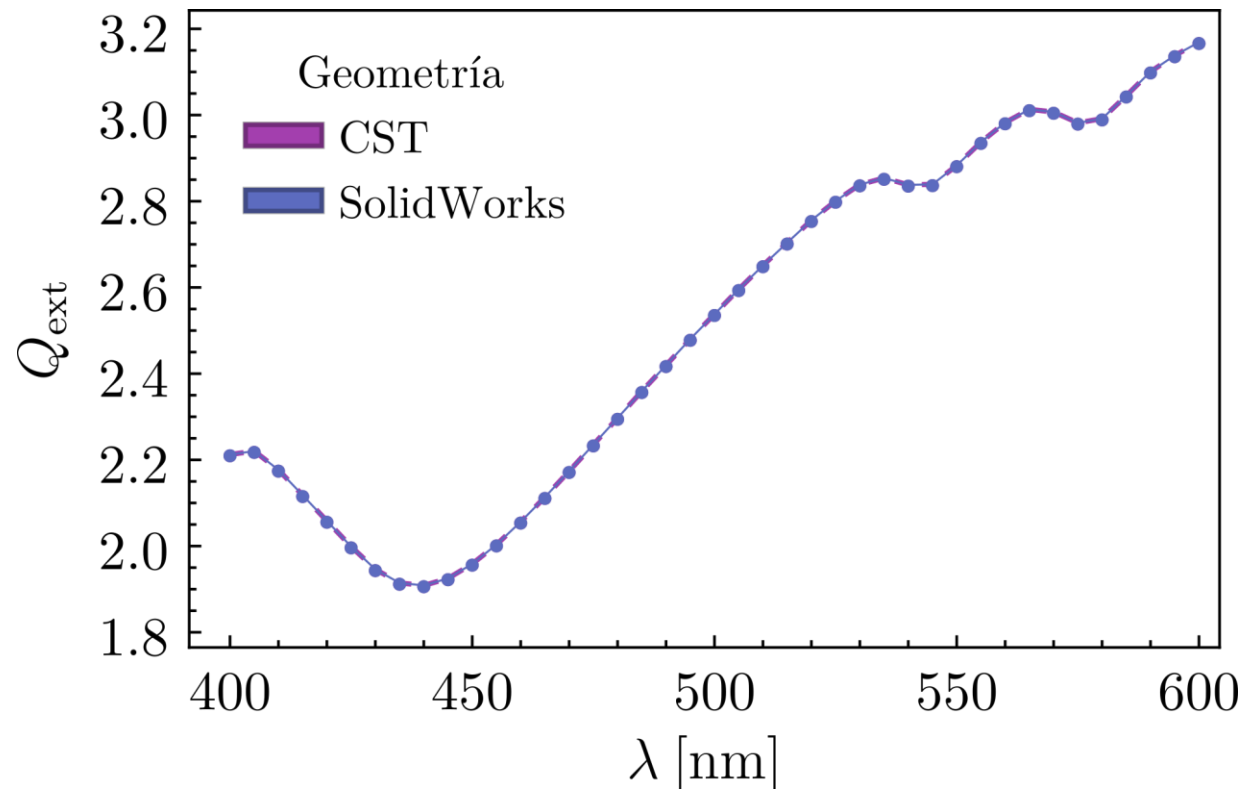


Convergencia numérica



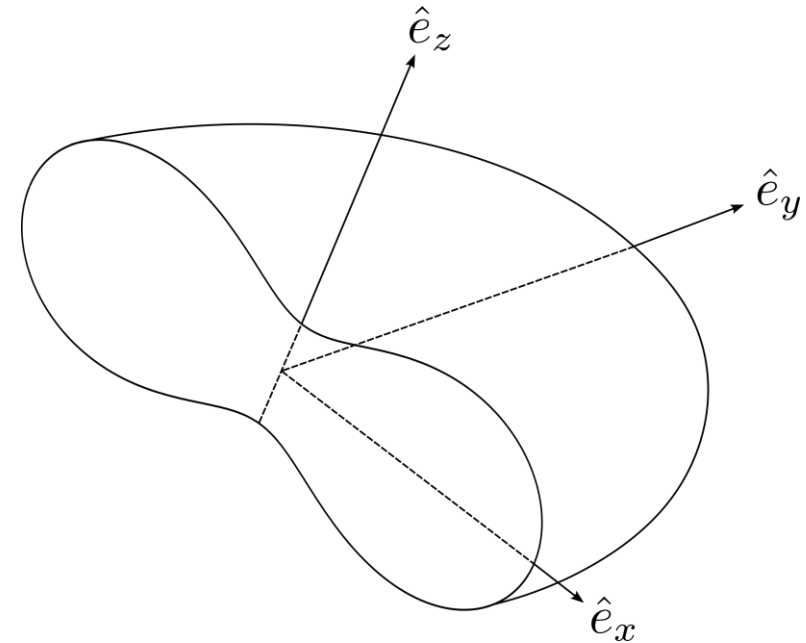
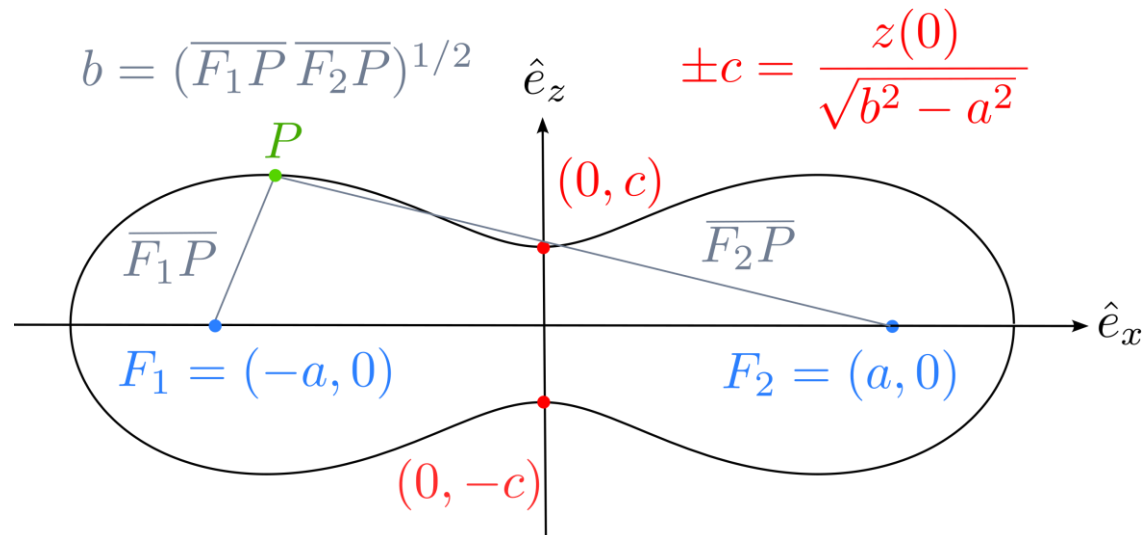
Efecto del mallado

- No se encontraron diferencias entre la geometría realizada en CST y la geometría importada desde SolidWorks

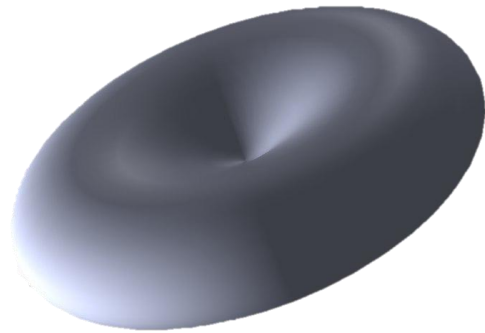


Óvalos de Cassini

$$z = \pm c(-a^2 - x^2 \pm (4x^2 a^2 + b^4)^{1/2})^{1/2}$$



Eritrocito sano



$$a = 2.6 \mu\text{m}$$

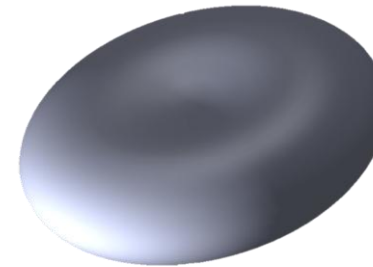
$$b = 2.7 \mu\text{m}$$

$$c = 0.8 \mu\text{m}$$

$$\text{Radio} = 3.75 \mu\text{m}$$

$$\text{Volumen} = 80.63 \mu\text{m}^3$$

Microcito



$$a = 2.4 \mu\text{m}$$

$$b = 2.7 \mu\text{m}$$

$$c = 0.5 \mu\text{m}$$

$$\text{Radio} = 3.62 \mu\text{m}$$

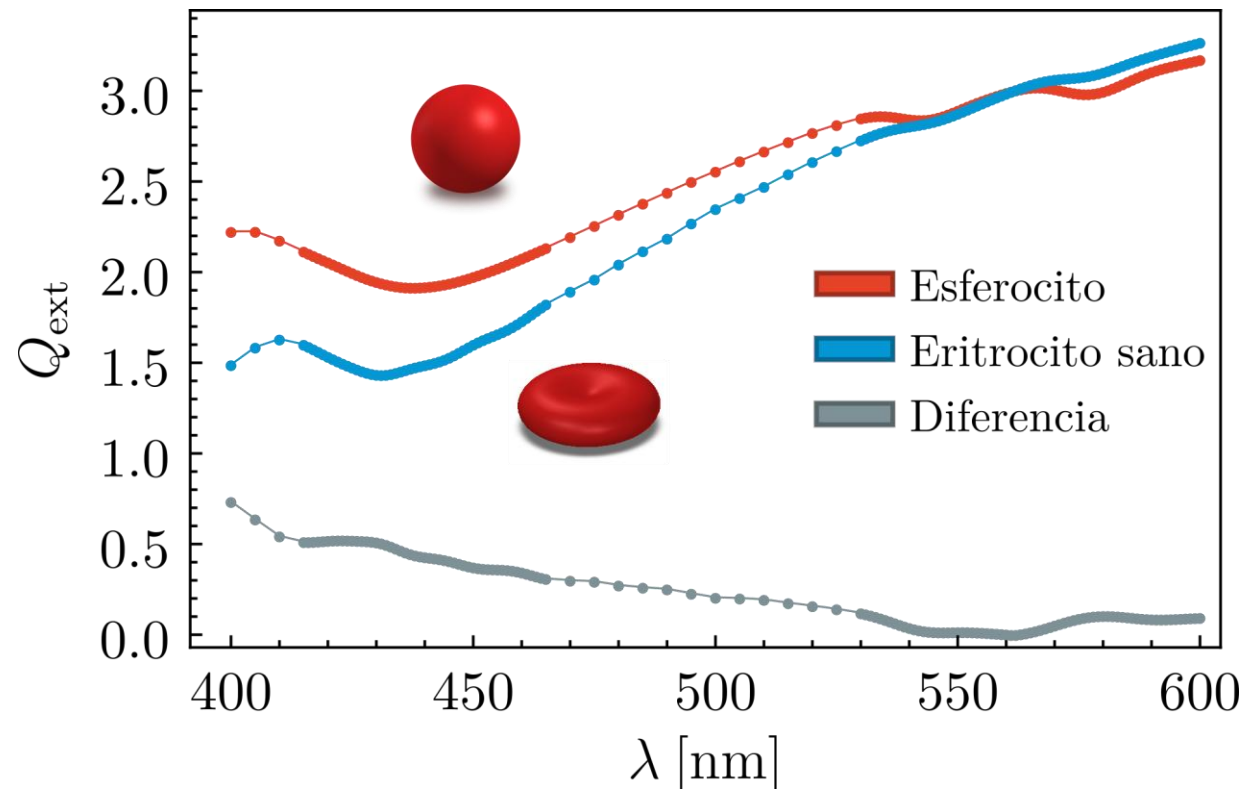
$$\text{Volumen} = 70.40 \mu\text{m}^3$$

Esferocito

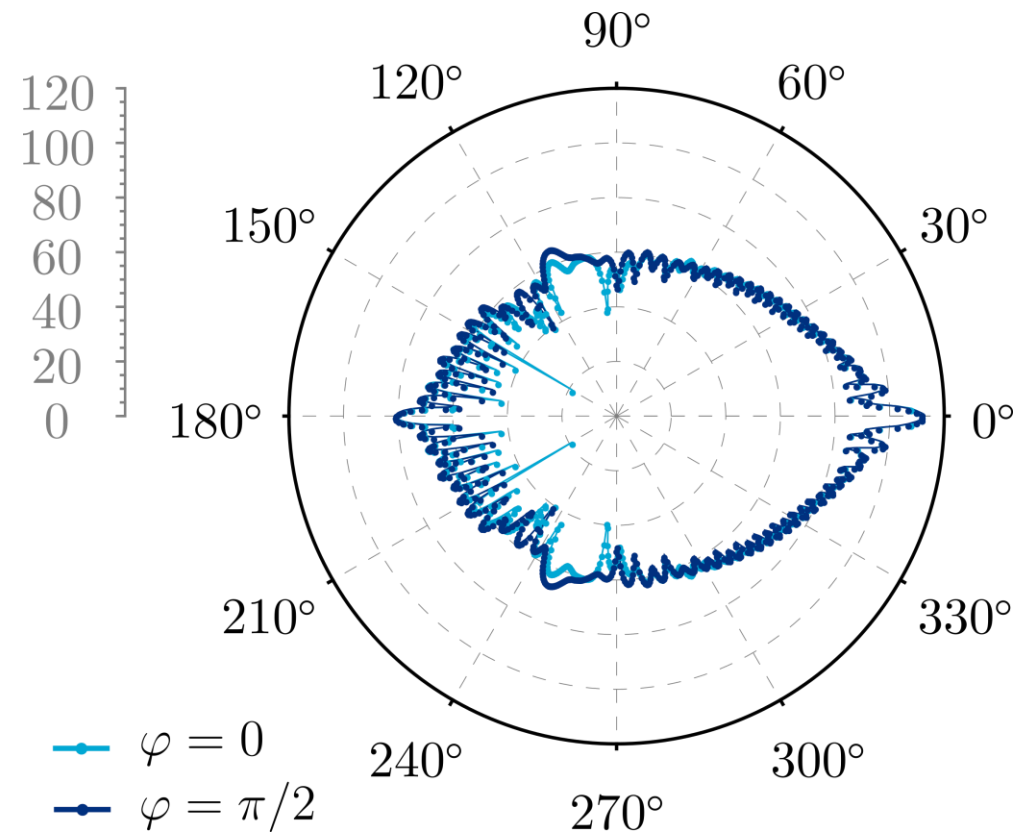
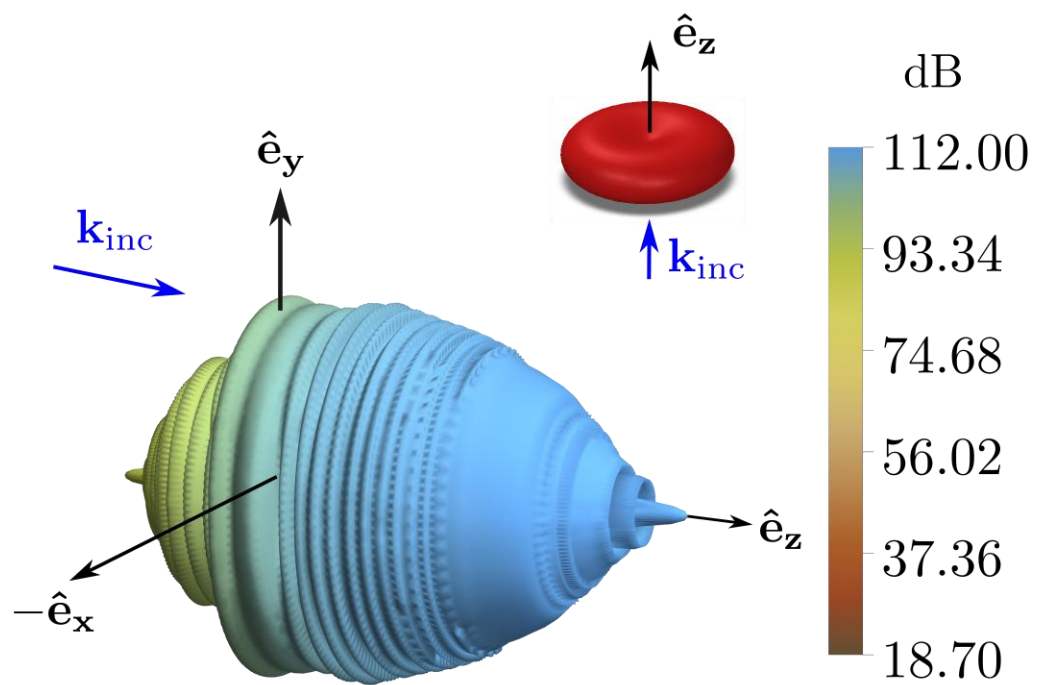


$$\text{Radio} = 2.68 \mu\text{m}$$

$$\text{Volumen} = 80.63 \mu\text{m}^3$$

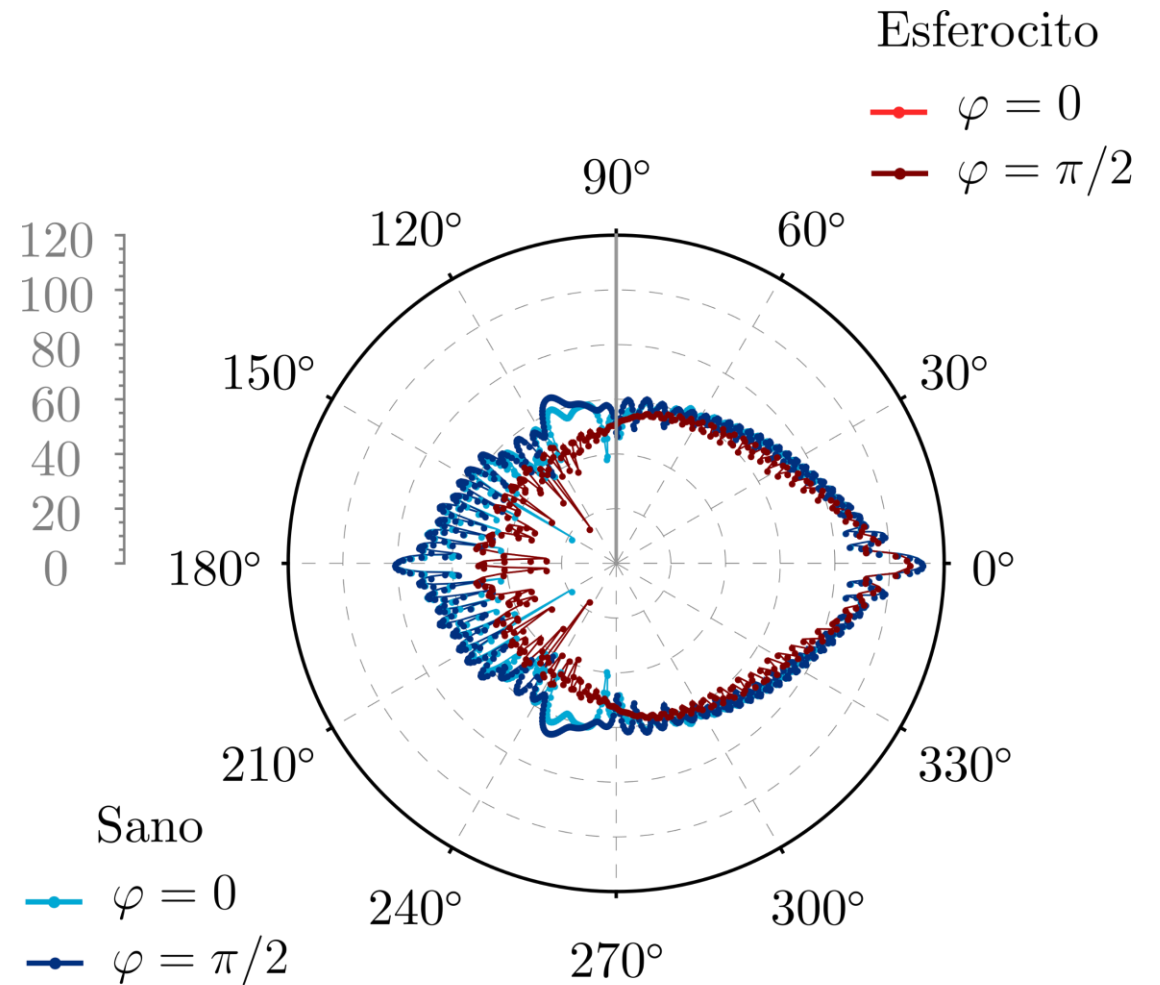
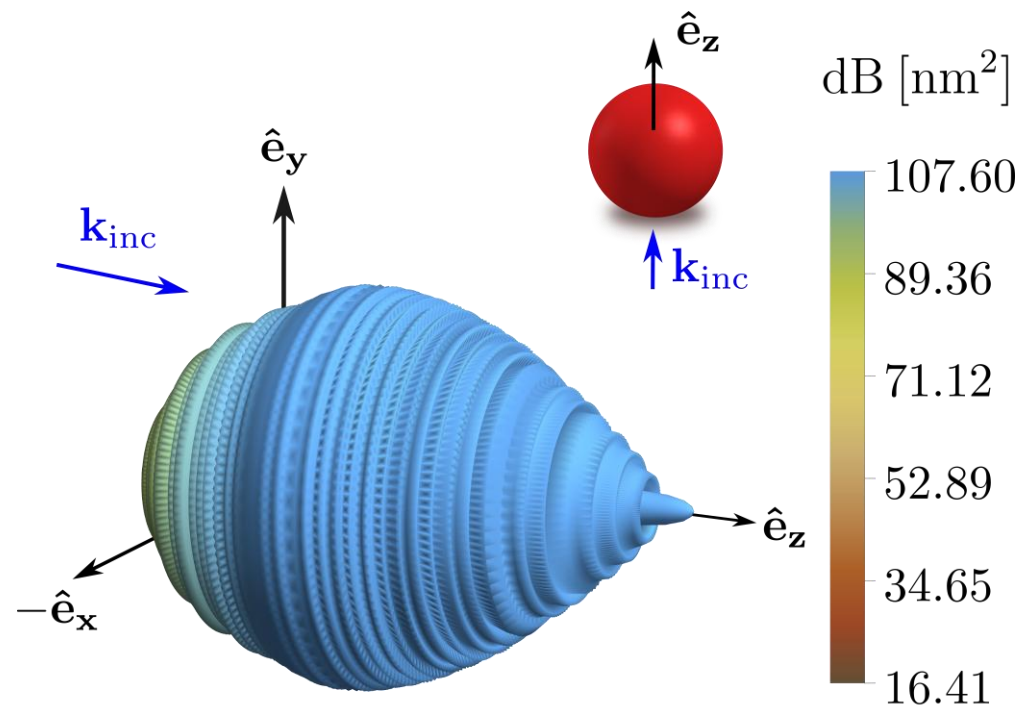


Eritrocito sano



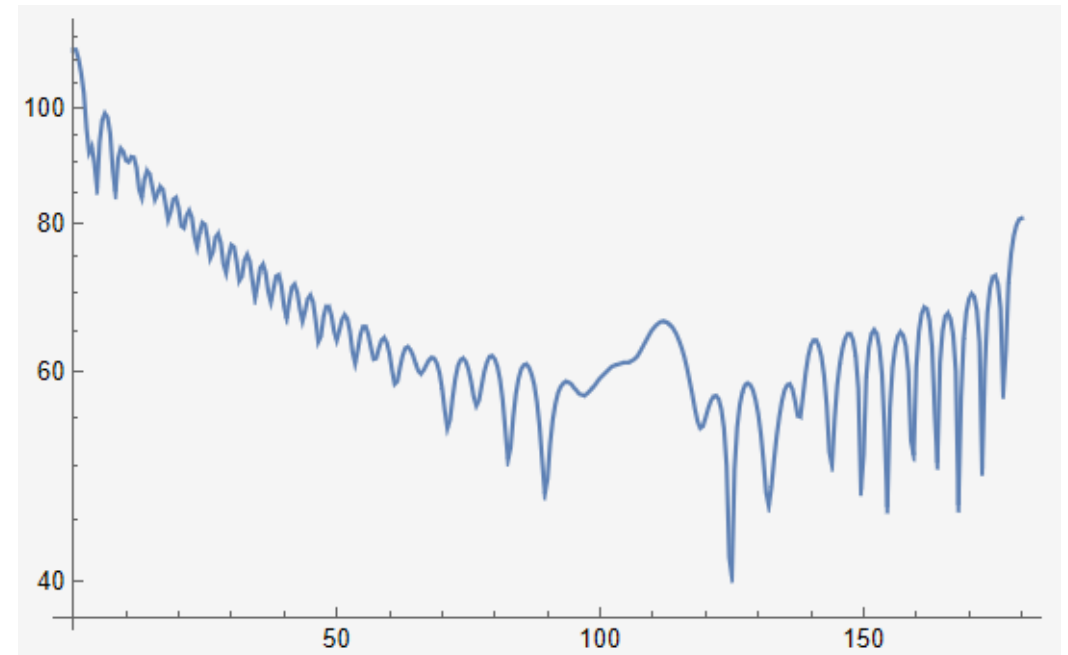
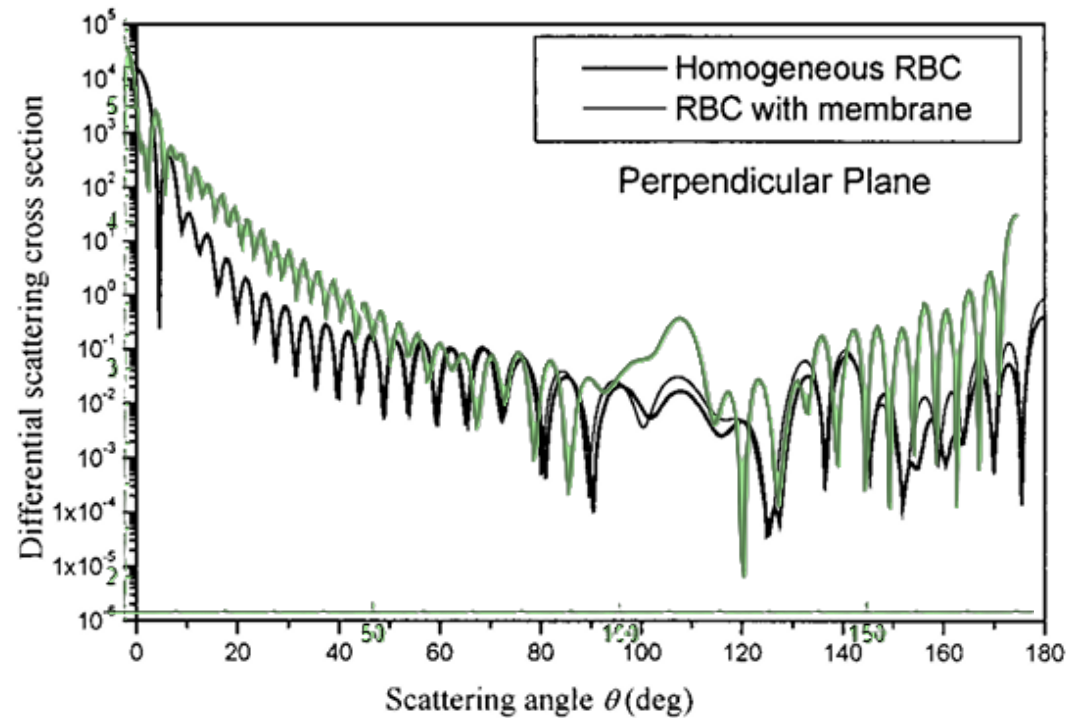
Esferocito

Campo lejano



Eritrocito sano

Campo lejano



Muchas gracias